

L2.8 Antiderivatives

§3.9 — run the rules backwards, then pin down the constant

Every answer with a $+C$ in it can be checked the same way: differentiate what you wrote and see whether you get back what you started with.

1 · The reverse power rule

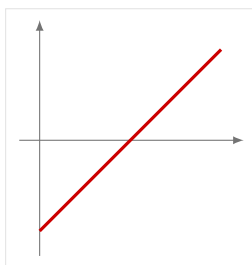
1. Find the antiderivative of each. **Rewrite every term as a power first.**

(a) $3\sqrt{x} - 2\sqrt[3]{x}$ (b) $(x - 5)^2$ (c) $\frac{10}{x^9}$

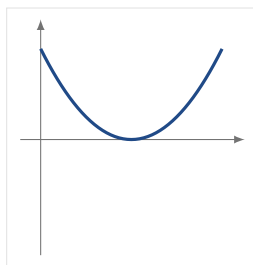
Stop. There is no reverse chain rule and no reverse product rule. Part (b) has to be expanded before any rule applies.

2 · Reading an antiderivative off a graph

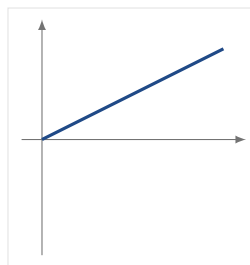
2. The graph of f is on the left. **Which of the three graphs beside it is an antiderivative of f , and how do you know?** Say what rules out each of the other two.



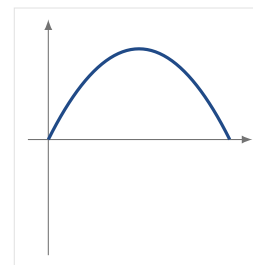
the graph of f



(a)



(b)



(c)

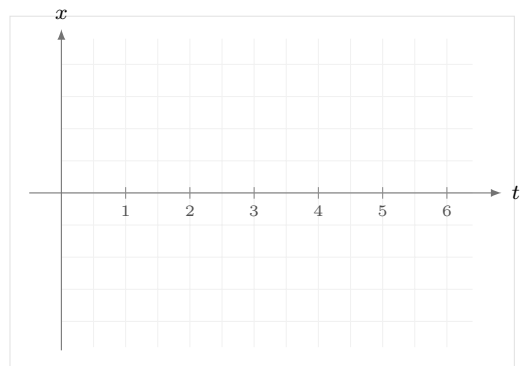
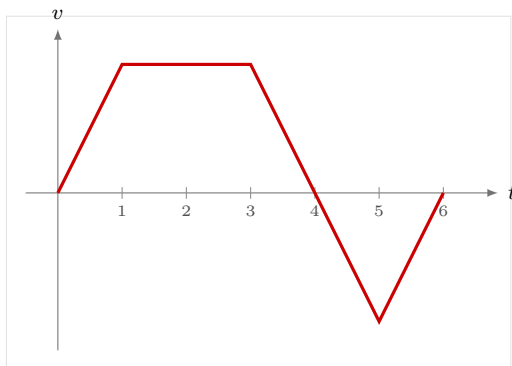
3 · Initial value problems

3. $f''(x) = 12x^2 - 6x + 2$, with $f'(1) = 5$ and $f(1) = -1$. Find f .

Stop. Two antidifferentiations, two constants. Resolve the first one *before* you antidifferentiate again.

4 · Position from acceleration

4. The velocity of an object is graphed below, and the object starts at $x(0) = 0$. **Sketch its position** on the empty axes.



Mark the t where x is largest, and say how you knew.

5. An object moves with **constant** acceleration a . Its velocity at time 0 is v_0 and its position at time 0 is x_0 . Show that

$$x = \frac{1}{2}at^2 + v_0t + x_0$$

5 · At the boards — antidifferentiation formulas

Start the period here, at the boards, before anything else on this sheet.

One antiderivative of $\sec^2 x$ is $\tan x$, because $(\tan x)' = \sec^2 x$. A *different* antiderivative of $\sec^2 x$ is $\tan x + 5$. Do you see why?

Even though you don't have a systematic method for finding antiderivatives, you can use your knowledge of differentiation, along with your inherent cleverness, to find some antidifferentiation formulas **by yourself**. Try to find antiderivatives of the functions below. You can always check your work by differentiating your answer.

6. **BOARDS** An antiderivative of $\cos x$ is:

7. **BOARDS** An antiderivative of $5 \cos x$ is:

8. **BOARDS** An antiderivative of $\cos(5x)$ is:

Stop. Differentiate whatever you write down. If it does not come back to $\cos(5x)$, fix it and check again.

9. **BOARDS** An antiderivative of $x^2 \cos(x^3)$ is:

10. **BOARDS** An antiderivative of $3x^2 \sin x + x^3 \cos x$ is:

11. **BOARDS** An antiderivative of $\frac{3x^2 \sin x - x^3 \cos x}{\sin^2 x}$ is: